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Data Science Fundamentals

knn

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Before we get started…

Practicalities

Learning objectives

- You understand and apply different types of regression, know when and how to use them, can execute them and put them in to practice as well as interpret the results.
- You recognize overfitting in regression models and learn techniques to prevent overfitting.
- You learn about different clustering techniques and understand when and how to use them for grouping data.
- You gain an understanding of dimensionality reduction techniques and learn how to apply them to compress and visualize data.

Course material

- **Canvas**
  - Slides & exercises from classes
- **Online**
  - https://www.w3schools.com/datascience/
- **Software**
  - JupyterLab (or work via Anaconda/VS Code if you prefer)
- **The Internet**



Course breakdown

- **Course Structure:**
  - Lectures: 3 hours per week (theory and exercises)
  - Attendance: Mostly not compulsory, but strongly recommended.

- **Examination Breakdown:**
  - 30%: Continuous assessment (mandatory excercises)
  - 70%: Final exam (oral, mostly theoretical with some practical elements)

Is it difficult to pass?

It’s perfectly doable **if you put in the work**

- Come to class
- Participate in the lessons
- Make your exercises



How to be an ideal student

- **Attend classes** - even when you party hardy
- **Show respect** - towards your instructor and fellow students
- **Be punctual** - or you'll find the door closed
- **Meet deadlines** - late submissions will earn a solid zero
- **Have a question?** - just ask! I don't bite
- **Know it all?** - help your classmates, they might repay you with pizza
- **Do your own work!** - shortcuts will lead to a short academic career ;)

Syllabus

Week - 1	Introduction & Python/JupyterLab Setup	Week - 9	Easter Holiday
Week - 2	Regression - Linear regression	Week - 10	Easter Holiday
Week - 3	Regression - Polynomial regression	Week - 11	Clustering - Hierarchical clustering
Week - 4	Classification - Logistic regression	Week - 12	Dimensionality reduction: Feature Analysis - PCA
Week - 5	Classification - Decision trees	Week - 13	Regularization: Lasso regression
Week - 6	Classification - kNN	Week - 14	Recap: Theory + Excercises
Week - 7	Clustering - kMeans	Week - 15	Exam Preperation + Q&A
Week - 8	Lifesnaps guidance		

(subject to change)

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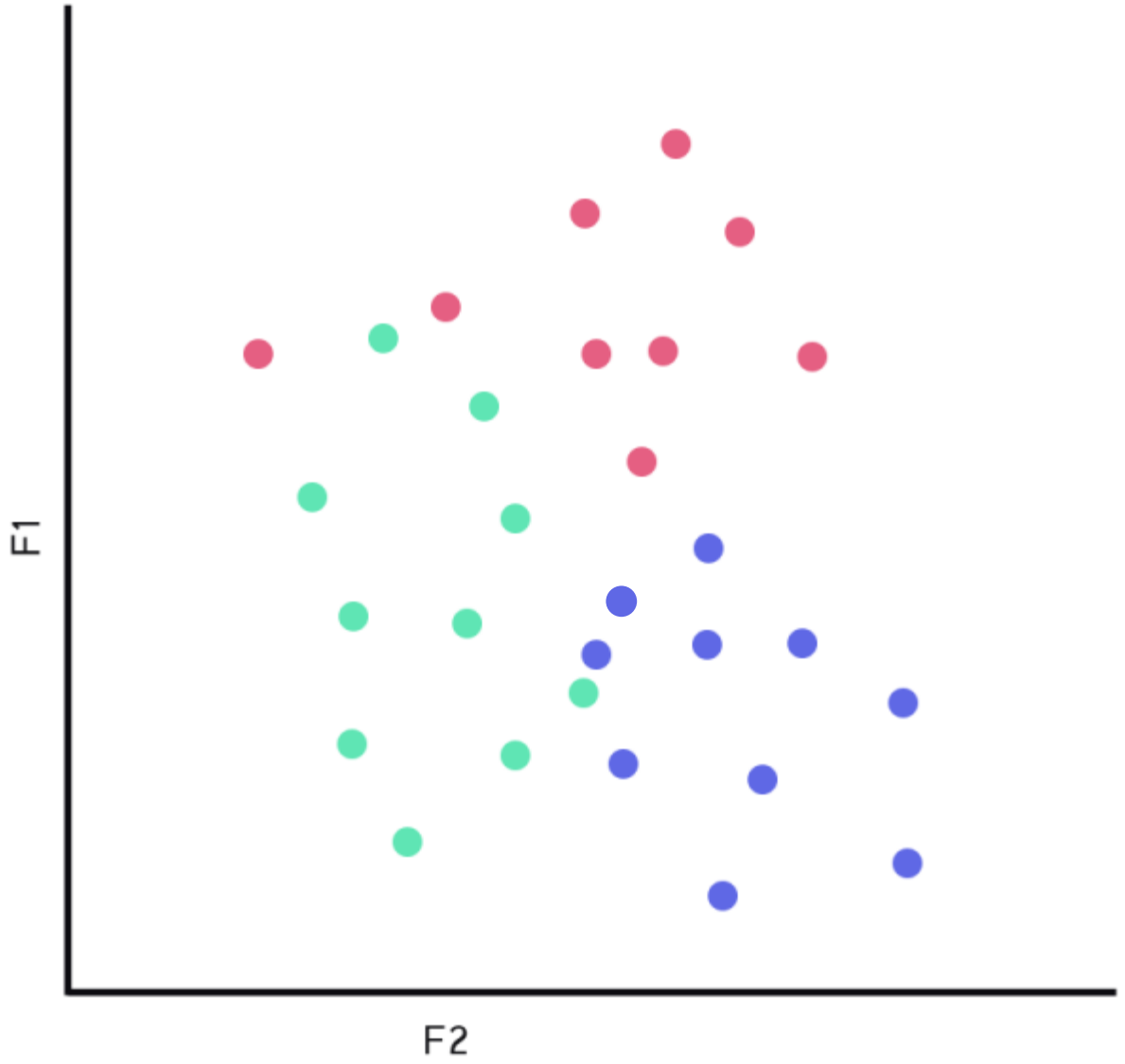
Now we get to the fun part

knn (k Nearest Neighbours)

What is knn

- ‘k nearest neighbours’
- Finds similar datapoints
- Each datapoint ‘votes’ on a class
- Majority vote: class with most votes wins!
- ‘k’ represents the amount of closeby datapoints to consider




F1
F2

Metrics

Once more, the same metrics apply:

- Accuracy
- Precision
- Recall
- Specificity
- F1 score

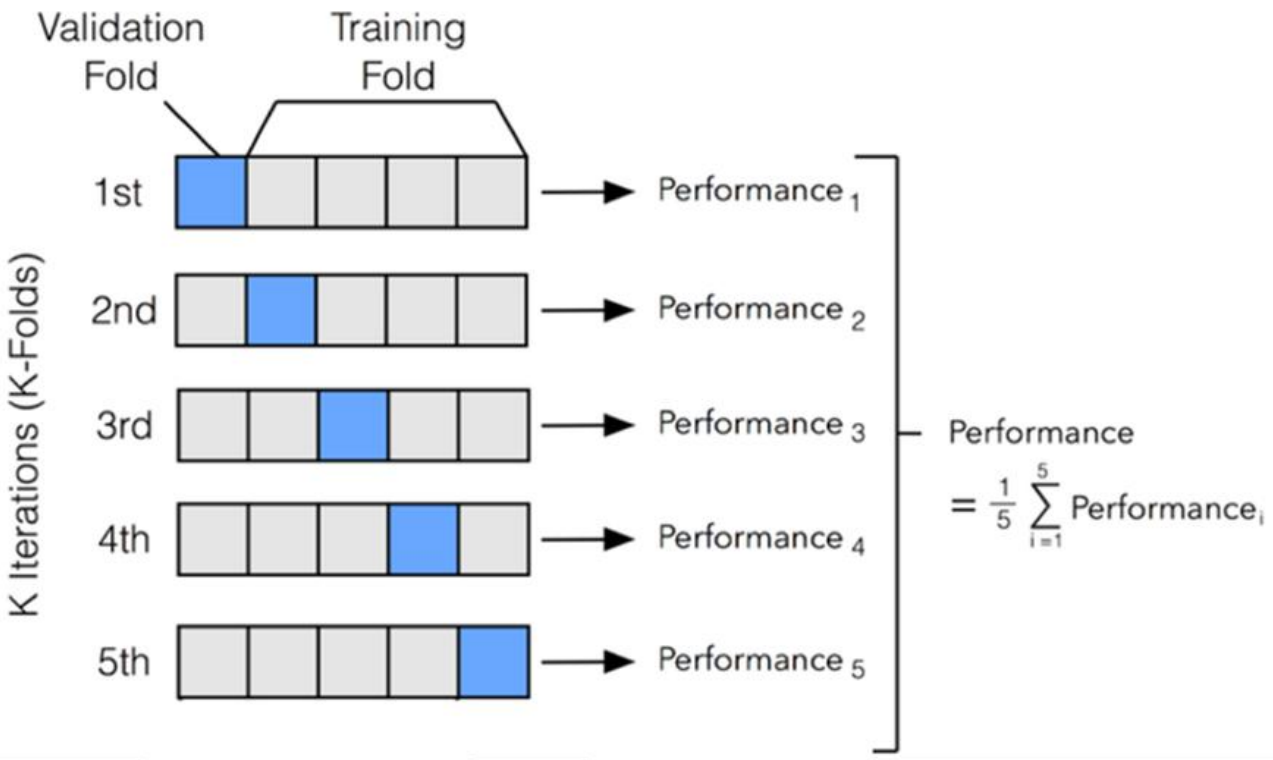


Overfitting

- ‘knn’ tends to overfit on low values of k.
  - → doesn’t learn patterns, just remembers some data
- On high values it tends to underfit
  - → doesn’t learn patterns, just does the same for every datapoint
- The best value is somewhere in between!
- How do we combat this?
  - Just like with random forests, we want to introduce **randomness** and **quantity** to get the best unbiased result

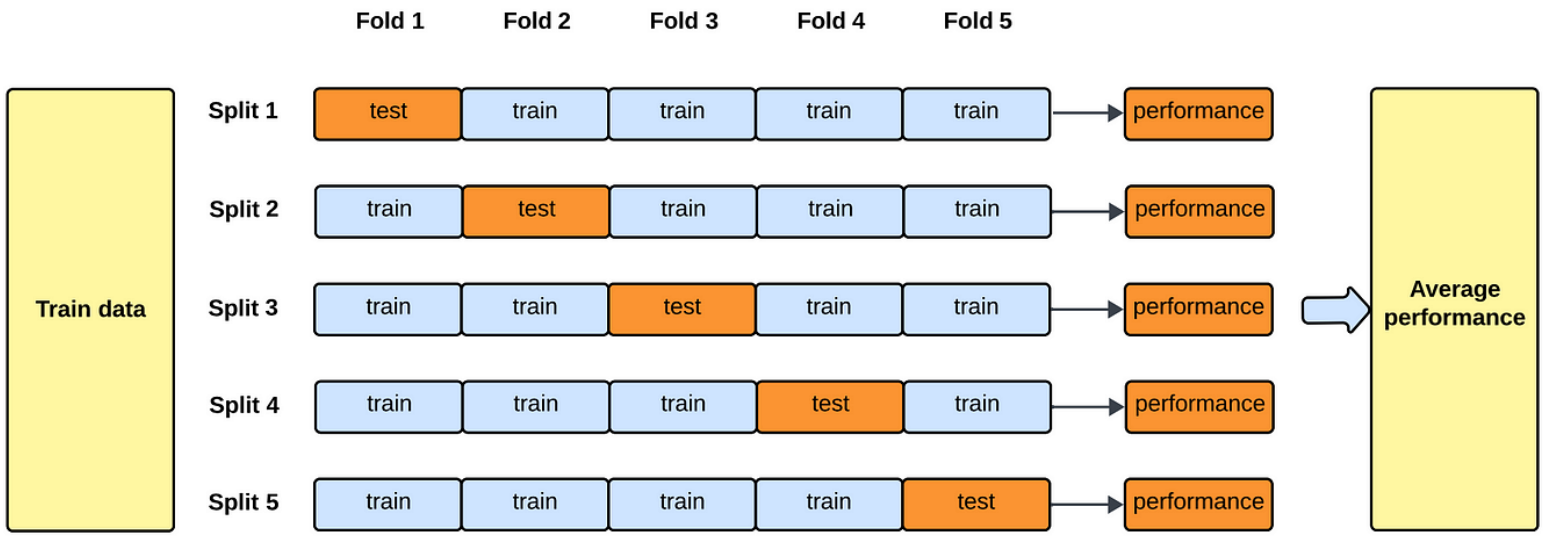
Cross validation

- Used to compare different models
- Create ‘folds’ in the data
- For each fold, train a model with that fold as test set
- Take the average outcome: classification → majority vote!
- Special cases:
  - ‘Holdout’
  - ‘leave-one-out’
  - ‘Stratified’



K-fold

- ‘k’ represents the amount of folds
- Most common model: 10-fold
- The value of ‘k’ cannot be higher than the amount of datapoints!



Source:Haden Pelletier, How-To: Cross Validation with Time Series Data (Medium 2023)

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Let's jump into action

Excercises

First, let's explore this together

- Go to canvas and download the notebook "DSF 06 – knn Notebook"
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Follow along!



Now it's up to you!

- Go to canvas and download the notebook "DSF 06 – knn Exercise"
- Also download the csv file called "Sonar.csv"
- Place them in the same directory
- Open the notebook in Jupyter
  - Open a CMD and run "jupyter lab"
  - Or launch it via anaconda
- Go through the notebook and make the excercises



I don't bite, ask me any questions you have!

And yes, this one is graded ;)